



UNDERSTANDING WEB3:

The Five Use Cases that will Dominate the Next Market Cycle

Published by Radix Publishing Limited

Web3 refers to a new layer of products and services for the internet built with decentralized networks, cryptography, tokens, smart contracts, and public ledgers.

With these technologies, Web3 has the potential to create:

1. an automated rules-based financial system existing entirely on the internet - **DeFi (including NFTs)**;
2. a way to fully control your identity, data, and privacy online - **Sovereign Identity**;
3. novel ways to organize and coordinate online communities and endeavors - **Decentralized Autonomous Organizations (DAOs)**;
4. ways to enrich **Gaming & Metaverse** experiences with finance, property rights, and identity;
5. the ability to find, connect, communicate and share with others outside of the control of any one party - **Decentralized Social**.

Perhaps Himgajria **summed it up best** as “where Web 2.0 was read-write, Web 3.0 is read-write-own”.

This exclusive report unpacks the problems that each of these use cases solves and elaborates on the opportunities for the next wave of disruption on the internet.



DeFi (including NFTs)

Decentralized Finance, DeFi, is a marketplace of financial products and services built out of tokens, NFTs, and smart contracts (self-executing computer code) on top of public decentralized ledgers, existing entirely on the internet.



Any aspect of finance, including payments, lending, investments, trading, or insurance, can be created in DeFi.

Harnessing the superpowers of tokens and smart contracts, DeFi has the potential to be a more efficient, open, and innovative financial infrastructure than the one we have today.

That's why DeFi is unquestionably Web3's most compelling use case. After all, there's no addressable market greater than the \$400 trillion of global finance.

Financial Products & Services

E.g. Payments, Lending Investments, Trading or Insurance

Decentralized
Applications
(dApps)

Tokens

Smart
Contracts

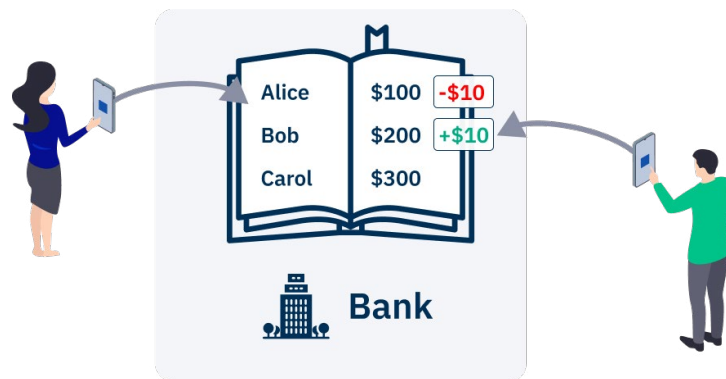
Public Decentralized Ledger

The Problem Today

Outside of physical cash, the financial system relies on intermediaries - such as banks - to hold records of various forms of money in their computers. Those records are called ledgers. Every time someone transacts, one or more ledgers are updated.

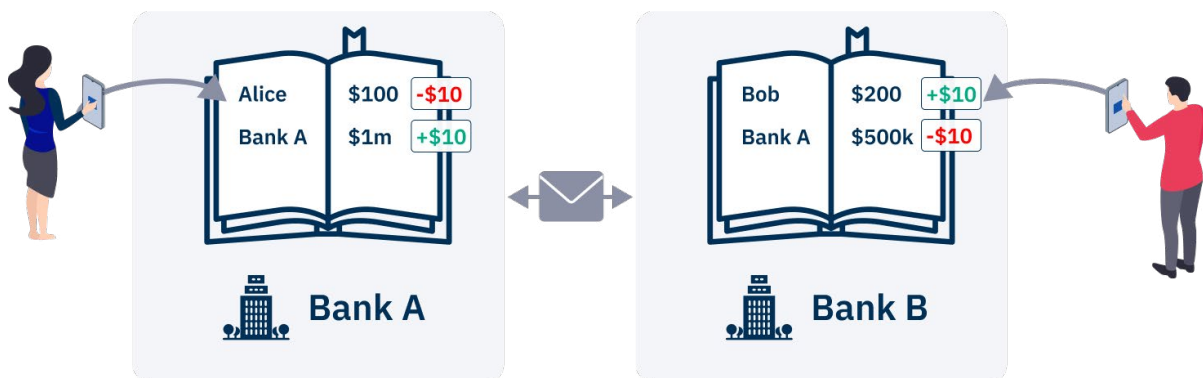
Through computers and manual processes, financial intermediaries enforce rules on which transactions are valid and when those ledgers can be updated.

If you wanted to pay a friend who banks at the same bank, after authenticating and checking you're authorized for your account, your bank updates one of its ledgers and the payment is settled.

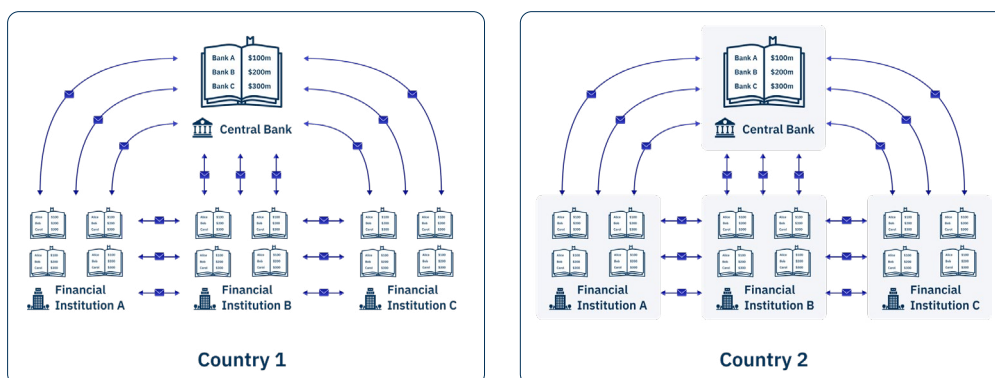


Alice sends \$10 to Bob

If you wanted to send money to someone at another bank, let alone another country... you can't. Contrary to popular belief, the only thing that can be sent are messages.



Alice sends \$10 to Bob at a different bank. If a transaction involves more than one intermediary, messages must be sent coordinating ledger updates across intermediaries.



Financial intermediaries process transactions through complex chains of messages and ledger updates. Those ledgers don't just record cash, but any form of financial instrument such as stocks, bonds, loans, or derivatives.

The financial system today is composed of thousands of these intermediaries updating their ledgers, sending millions of messages to one another every day.

This creates two problems:

1. Finance is Inefficient:

- **Cost** - Financial intermediaries are expensive to run. They have thousands of people, processes, and technology to pay for, to process transactions and ensure their ledgers are accurate.
- **Profit & Rent** - Intermediaries are profit and rent-seeking. If they can, they will extort their customers. Why are merchants charged up to 3% for card payments? Why does it cost 10% to send money overseas? Why do you get only 0.1% interest on your current account?
- **Liquidity & Settlement** - By fragmenting global finance over thousands of private ledgers, liquidity (the ability for money to move) is slow. This is why it sometimes takes a week for an international payment to settle, because each “hop” between financial intermediaries takes time.

2. Finance is Closed and Uninnovative:

- **Transparency** - The private ledgers of financial intermediaries are opaque. It's not possible to see what's going on inside. This is one reason why we had the 2008 financial crisis, as no one knew what was going on.
- **Trust & Permissions** - Intermediaries have to be trusted and can deny who can access their services. Not everyone has equal access to financial services. Banks in places as varied as [China](#), [Canada](#), and [Lebanon](#), have over 2021 and 2022 frozen individuals' access to their money.
- **Competition & Innovation** - Intermediaries close off their technology and are anti-competitive. They have proprietary product stacks that are closed off to third party contributions.

The Opportunity in Web3

If you think about it, you can boil finance down to two things:

- numbers that represent **assets** of value, such as cash or other financial instruments recorded in ledgers;
- **rules** that determine who is allowed to change those numbers and in what ways.

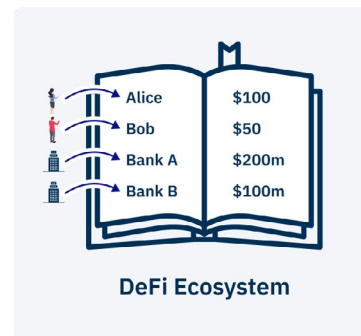
And DeFi does exactly this, using:

- **tokens and NFTs** to represent anything of value recorded on public decentralized ledgers;
- self-executing **smart contracts** that enforce rules on who is allowed to transact with those tokens and in what ways.

The key difference then is that in traditional finance the representations of value are held in thousands of **private centralized ledgers**, with rules enforced by each intermediary. Whereas in DeFi, tokens and NFTs are held on **public decentralized ledgers**, with everyone having equal access, with rules enforced automatically by self-executing smart contracts.

DeFi is thus a public good. A shared computing environment that no one party has control of that anyone can use to create their own representations of value (tokens) and create their own sets of automatic rules on who is allowed to transact with tokens and in what ways (smart contracts). Anyone can create their own financial application, no matter how small the use case.

This solves those two problems because:



DeFi - everyone has equal access to the same ledger

1. DeFi is efficient:

- **Cost** - DeFi, through smart contracts, is automated. Billions can be saved by providing financial products and services without the need for costly processes and entities.
- **Profit & Rent** - DeFi's peer-to-peer and peer-to-pool relationships disintermediate the ability for institutions to extract profit and rent.
- **Liquidity & Settlement** - As a shared computing environment, DeFi transactions are global, 24/7, and instant. No waiting around for some bank risk committee to approve or review a transaction. You can move \$1Bn in 15 seconds. Markets continue to operate whether it's the weekend or not.

2. DeFi is open and innovative:

- **Transparency** - DeFi's ledgers are public and transparent. Where you shine a light, corruption hides. Anyone can view the entire state of the financial system. This makes regulation of the system as a whole easier and more robust. Public and transparent data allows for better capital and risk allocation.
- **Trust & Permissions** - In DeFi, through use of cryptography, you no longer have to trust an intermediary to hold your assets for you - you do so yourself, directly on the public ledger: DeFi is trust-minimized. In DeFi, anyone can access any financial product around the world. Anyone can contribute through open source code: DeFi is permission-minimized.
- **Competition & Innovation** - Because of this, anyone can create or hold their own token that can represent anything, build their own dApp, compose with what's built before, and collaborate in ways never before seen, with lower barriers to entry and instant transparency and trust. This should result in an explosion in innovation and competition of the financial products and services available to everyone.

The Opportunity - What Still Needs to be Solved

But DeFi is not yet ready for prime time.

There are still a multitude of problems (and opportunities) to solve:

Hacks

While DeFi is automated through smart contracts, this is also perilous if something goes wrong, as exemplified by the over \$3 billion in DeFi hacks over 2021 and 2022. This is not the fault of smart contract developers as programming languages such as Solidity are incredibly difficult to make secure. Instead, we need a new type of programming language that treats assets such as tokens and NFTs as native first class resources, such as [Scripto](#).

Transaction Fees

While DeFi can theoretically provide a more efficient infrastructure for finance, today it can't because platforms such as Ethereum are not scalable. This creates a bidding war for transactions, pushing up their price. There is no way that the future of finance will be on a platform that can sometimes charge \$200 for a transaction.

Support

If something goes wrong in finance today, you pick up the phone. In DeFi, there's no one. There will be a need for centralized services that provide support or insurance to individuals who aren't quite ready yet to be their own bank.

Overcollateralization

DeFi today is overcollateralized. That means that any financial activity that involves borrowing is typically supported by a deposit that is greater than the amount borrowed. While this is fantastic for risk management - definitely needed given DeFi's pseudonymous nature - it also makes real world lending difficult. Identity and risk scores might go some way to solving this.

Privacy

DeFi's openness is also a curse. With every transaction public, privacy is diminished. DeFi will need to find ways to do this, through perhaps privacy preserving techniques such as zero-knowledge technologies, or use centralized intermediaries to obfuscate the person behind transactions on the public ledger.

Financial Crime

While DeFi's permission and trust-minimization fosters competition and innovation, it also means anyone can set up their own scam. This also means anyone can transact - whether you're a terrorist, money-launderer, or sanctioned. For DeFi to be the future home of finance, it will need ways to combat these risks.

Sovereign Identity

Sovereign identity refers to the concept that you are in control of your own identity online. No one can pretend to be you. No one can take your identity away from you. You can take it anywhere you want to go. You are sovereign.



The Problem Today

It comes as no surprise to anyone that the current mechanisms for authentication and identification on the internet are broken.

How many separate user accounts do you have today? With how many passwords? Do you reuse those passwords? Even if you use a password service, such as 1password, are you still in control, or are you still relying on someone else? How many of your passwords have been hacked or stolen out of the [24 billion](#) or so that are available on the dark web?

As our physical lives become ever more entwined with our digital, improving internet users' ability to control their sovereign identity becomes ever more important.

Google, Apple, and Microsoft are of course [not sitting still](#), with the race on to make the internet a password-less place.

But schemes such as this are still reliant upon these megacorps to safeguard your identity for you. You are not in control. They are.

The Opportunity - Web3

Web3 offers a decentralized alternative to entrusting third parties with your entire digital life. It does this through two key mechanisms:

- **Web3 Wallets**, leveraging cryptography, allow internet users to prove their identity online without needing to trust or rely upon any central authority.
- **Decentralized networks**, leveraging token economics, create financial incentives to process and store immutable records of data.

This allows for any kind of application to be created, with permissioning systems of any complexity, leveraging this combination of identity, data storage and smart contracts.

Why is this Beneficial?

- Instead of relying on a circus of websites to store your username and passwords, with Web3, you just identify yourself with your wallet.
- Instead of relying on them to store your data, some of which could be private, you instead store it in an encrypted form on a decentralized network such as IPFS or Arweave.
- Given you are in control of your identity - you can take it with you. It is portable and reusable across any number of services. You are not trapped in a single ecosystem.

Exciting technologies such as zero-knowledge proofs allow for services to be built that can prove that a certain statement derived from a piece of data is true without needing to reveal the underlying data itself. For example, a user can prove that they are over a certain age without needing to reveal it. You can prove to an insurer that you have not had a particular disease without needing to reveal your entire medical history.

Lastly, through the use of tokens such as NFTs, these could be leveraged to create systems of authorization. An NFT could for example provide access to a private Discord server. If the user who holds it no longer needs it, they could sell it to someone who does.

The Opportunity - What Still Needs to be Solved

Whilst proving your identity using Web3 techniques remains almost free, storing data on decentralized networks is not.

Because decentralized networks replicate data hundreds or thousands of times across many nodes, it is far more expensive to store data in a trust and permission-minimized manner than the equivalent today in Web2. Some dApps that attempt to recreate today's services in an entirely decentralized way will likely see only a small subset of the most sovereign-conscious users use their services.

What's more likely are hybrid models utilizing wallets for identification without passwords, and a combination of apps that are built that leverage centralized hosting solutions for the majority of data requirements, and decentralized Web3 solutions for only the most sensitive or important data.



Decentralized Autonomous Organizations (DAOs)



A Decentralized Autonomous Organization (DAO) is a new way of coordinating activity on the internet that leverages tokens to create shared treasuries and systems of economic incentives; smart contracts to create rules that stipulate how and when those tokens can be used; and methods of communication to allow communities to discuss and vote on matters of interest.

As a result, DAOs allow for new ways of collaborating globally online without needing to utilize the traditional banking and legal systems.

The Problem Today

If you set up a project today and want to have some means of rewarding those that contribute to it, you will likely need to have a bank account. If you'd like to do so in a way that isn't entirely your personal responsibility, you will likely need to set up a legal entity - often with other shareholders or directors. Both of these things require a country to be based in. Both need a minimum amount of financial and legal resources to set up.

This can prove problematic if the project is small or just an experiment, or even worse, cross border. When larger, a small subset of individuals often still retains legal and financial control of the company in whatever country it's based in.

Company finances and decision-making processes can be opaque.
Cross-border payments can be slow and expensive.

The Opportunity - Web3

DAOs go some way to solving this.

First, they allow a decentralized community of anyone on the internet to pseudonymously contribute or raise capital (tokens) to shared treasuries with no one party in control.

Second, leveraging smart contracts and tokens, governance and incentive systems can be created that determine when the funds from the treasury can be disbursed. These governance systems typically involve some means of voting, weighted in a variety of ways. The systems of incentives on the other hand are designed to encourage behaviors and activities that benefit the goals of the project, such as for completing a task or providing liquidity for a certain endeavor.

Third, DAOs often have some means of allowing its community to communicate, such as through chat or forums.

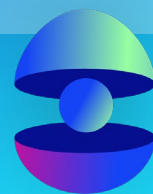
Why is this Beneficial?

- Because DAOs (and DeFi) exists entirely on the internet, the complexity of cross-border finance and legal arrangements can be reduced.
- Setting up a DAO also costs next to nothing as compared to the equivalent for a legal entity, lowering barriers to entry and allowing for the formation of groups online that otherwise wouldn't be able to exist, especially since participants can stay pseudonymous.
- The public nature of decentralized ledgers also offers a glimpse to future organizations that are radically transparent and perhaps more resilient - eschewing any overdependence on any one individual or entity.

The Opportunity - What Still Needs to be Solved

DAOs are still incredibly new and so different governance and incentive mechanisms are still very much in experimentation mode. Some systems have resulted in concentration of power and over-reliance on a small subset of individuals to do the lion's share of work. Some have resulted in sclerosis - where too many voices means nothing gets done.

DAOs represent an exciting new paradigm for organizing human activity. But much research, experimentation and maturing is still required before large swathes of human endeavor falls under their sway.



Gaming and Metaverse

Games and metaverses are not new to Web3.

We've had online communities building, playing, and sharing their gaming and other rich online experiences for years.

What's different now is that DeFi, Sovereign Identity, and DAOs add an additional layer beneath the front end of the game or metaverse experience. One that enriches gaming & metaverse experiences with finance, property rights, and identity.



The Problem Today

Current game and metaverse developers are beholden to the large platforms who take a cut in every piece of revenue and can deplatform any title with a moment's notice. Game and metaverse players are also held hostage. Spend \$100 on an in-game item? It's not really yours. Want to take your achievements with you when you change games? You can't. Want to form a collective to build new experiences for others with the friends you've made online? You'll need a bank account and legal entity if you want to truly scale it.

So how might Web3 improve gaming and metaverse experiences?

The Opportunity - Web3

None of the above issues are solved by any products or services that we haven't yet covered. The opportunity in gaming and metaverse is just the application of DeFi, Sovereign Identity, and DAOs to this particular corner of the internet:

- DeFi allows for game and metaverse participants to pay for the development of the games they play; for any in-game item, land or property being tokenized; and for smart contracts allowing for complex applications and organizations to form entirely digitally. The result of which should make gaming and metaverse experiences more immersive, encouraging virtual economies of competition and collaboration.
- Sovereign Identity allows for participants to truly own and control their experience. If you leave a game, now you can actually take your assets, achievements and friends with you, allowing for persistent communities to form across games and metaverses.
- DAOs allow for digital communities to form with much more impetus than before - finally able to harness systems analogous to money and law to achieve specified objectives.

The Opportunity - What Still Needs to be Solved

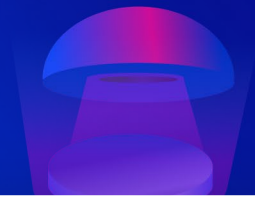
The products and services currently being developed for DeFi, Sovereign Identity and DAOs need to be tailored for gaming and metaverse audiences.

5

Decentralized Social

Every internet user, to varying extents, relies on social media of some sort to either stay in touch with friends, family, colleagues; or consume content such as the news.

Like Gaming and Metaverse, Web3 doesn't change any of this, at least on the surface. But through DeFi, Sovereign Identity, and DAOs, Web3 promises to initiate a movement that gives users back control of how they experience the internet.



The Problem Today

From influencing elections, censoring those whose views are not liked, through to losing or taking advantage of the private data they steward, the largest social media networks have outsized influence on society.

The Opportunity - Web3

Web3, through the same mechanisms as have already been described, has the potential to provide an alternative choice for social:

- DeFi allows for economic models for social that don't rely on advertising and selling user data.
- Sovereign Identity allows users to be in control of their accounts, privacy, and data, without relying on any central platform.
- DAOs allow for the coordination of these activities in ways that better reflect the interests of the users of the platform.

The Opportunity - What Still Needs to be Solved

Network effects. A social graph is only useful if it already has users. There will need to be compelling use cases if users are to switch wholesale from the current monopolies to a Web3 alternative.

Hosting and computing in decentralized environments is expensive. Replicating thousands of photos and videos is not efficient. Running complex news feed or recommendation algorithms on decentralized networks may not offer a comparable experience.

Conclusion

Web3 heralds the formation of an efficient, open and innovative rules-based financial system for the internet; offers individuals the ability to fully control their identity and data online; and gives online communities the cooperation-enhancing superpowers of property rights and law in a way that is entirely digital.

Consequently, as the products and services delivering Web3's use cases are built over the coming years, there is potential for a rewiring of the social fabric of the internet, and by extension, society at large. But this is very much only the potential to do so.

Web3 is still in its infant prototype mode, incapable of delivering the above promise. Three key problems that still need resolution are:

- **User experience** for Web3 is complex, buried in technical jargon, and quite frankly - extremely scary. Only the most tech-savvy are able to put up with and navigate Web3 today.
- **Developer experience** is poor. It typically takes two years to become even half-way competent in programming languages such as Solidity. Even then, we see billions of dollars of hacks every year as a result of even the smallest of mistakes from experienced developers.
- **Non-scalable infrastructure** means that fees for some transactions remain orders of magnitude higher than their equivalents in traditional finance today.

Radix will solve all of these problems with a radically different approach to all other Web3 infrastructures:

- A **revamped wallet experience** that will make Web3 far easier, safer, and intuitive to use.
- A **revolutionary developer experience**, with the asset-oriented Scrypto programming language and invention of the world's first "DeFi Engine", Radix Engine, taking away much of the complexity of building Web3 today.
- A fundamental **re-architecture of consensus and data** that will allow Web3 on Radix to scale linearly, like the internet itself, without ever compromising on the network's ability to combine transactions and applications together.

If you'd like to learn more about Radix:

- Start here: <https://www.radixdlt.com/brave>
- To keep up to date with everything relating to the Radix Public Network and ecosystem, be sure to jump into our core community channels [Telegram](#), [Discord](#), [Twitter](#) and [Reddit](#).
- The Radix [YouTube](#) channel showcases events and highlights interviews with global financial specialists.
- The [Radix Report](#) is a fortnightly update from the marketing & development teams at RDX Works.